

Exercise JA

Q1. Let $-\frac{\pi}{6} < \theta < \frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x \sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x \tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals

- (A) $2(\sec\theta - \tan\theta)$ (B) $2 \sec\theta$ (C) $-2 \tan\theta$ (D) 0

Ans. C

Sol. Roots of the equation $x^2 - 2x \sec\theta + 1 = 0 \rightarrow \sec\theta - \tan\theta, \sec\theta + \tan\theta$
 $\alpha_1 = \sec\theta - \tan\theta, \beta_1 = \sec\theta + \tan\theta$ ($\because \alpha_1 > \beta_1$)
 Roots of the equation $x^2 + 2x \tan\theta - 1 = 0 \rightarrow \tan\theta + \sec\theta, -\tan\theta - \sec\theta$
 $\alpha_2 = \sec\theta - \tan\theta, \beta_2 = -\sec\theta - \tan\theta$ ($\because \alpha_2 > \beta_2$)
 Hence $\alpha_1 + \beta_2 = -2\tan\theta$

Q2. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to :

- (A) $3 - \sqrt{3}$ (B) $2(3 - \sqrt{3})$ (C) $2(\sqrt{3} - 1)$ (D) $2(2 + \sqrt{3})$

Ans. C

Sol.
$$\sum_{k=1}^{13} \frac{\sin\left[\left(\frac{\pi}{4} + \frac{k\pi}{6}\right) - \left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right)\right]}{\sin\frac{\pi}{6}\left(\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)\sin\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right)\right)} = 2 \sum_{k=1}^{13} \left(\cot\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right) - \cot\left(\frac{\pi}{4} + k\frac{\pi}{6}\right)\right)$$

$$= \left(\cot\frac{\pi}{4} - \cot\left(\frac{\pi}{4} + \frac{13\pi}{6}\right)\right) = 2\left(1 - \cot\left(\frac{29\pi}{12}\right)\right) = 2\left(1 - \cot\left(\frac{5\pi}{12}\right)\right) = 2(1 - (2 - \sqrt{3}))$$

$$= 2(\sqrt{3} - 1)$$

Q3. Let α and β be non-zero real numbers such that $2(\cos\beta - \cos\alpha) + \cos\alpha \cos\beta = 1$. Then which of the following is/are true :

- (A) $\tan\left(\frac{\alpha}{2}\right) + \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$ (B) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right) = 0$
 (C) $\tan\left(\frac{\alpha}{2}\right) - \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$ (D) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 0$

Ans. A, C

Sol. $\cos\alpha = \left(\frac{1-a}{1+a}\right); \quad a = \tan^2\frac{\alpha}{2}$
 $\cos\beta = \left(\frac{1-b}{1+b}\right); \quad b = \tan^2\frac{\beta}{2}$
 $2\left(\left(\frac{1-b}{1+b}\right) - \left(\frac{1-a}{1+a}\right)\right) + \left(\left(\frac{1-a}{1+a}\right)\left(\frac{1-b}{1+b}\right)\right) = 1$

$$\begin{aligned}
2((1-b)(1+a) - (1-a)(1+b)) + (1-a)(1-b) &= (1+a)(1+b) \\
2(1+a-b-ab - (1+b-a-ab)) + 1-a-b+ab &= 1+a+b+ab \\
4(a-b) &= 2(a+b) \\
2a-2b &= a+b \\
a &= 3b \\
\tan^2 \frac{\alpha}{2} &= 3 \tan^2 \frac{\beta}{2} \\
\tan \frac{\alpha}{2} &= \pm \sqrt{3} \tan \left(\frac{\beta}{2} \right)
\end{aligned}$$

Q4. Let $f:[0,2] \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = (3 - \sin(2\pi x)) \sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right).$$

If $\alpha, \beta \in [0, 2]$ are such that $\{x \in [0, 2] : f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is

Ans. 1

Sol.
$$f(x) = (3 - \sin 2\pi x) \sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right)$$

$$\text{Let } \pi x - \frac{\pi}{4} = \pi y \Rightarrow x = y + \frac{1}{4}$$

$$f(x) = \left(3 - \sin 2\pi\left(y + \frac{1}{4}\right)\right) \sin y + \pi - \sin\left(3\pi\left(y + \frac{1}{4}\right) + \frac{\pi}{4}\right)$$

$$= \left(3 - \sin\left(2\pi y + \frac{\pi}{2}\right)\right) \sin y - \sin\left(3\pi y + \frac{3\pi}{4} + \frac{\pi}{4}\right)$$

$$= (3 - \cos 2\pi y) \sin \pi y + \sin(3\pi y)$$

$$= (3 - (1 - 2 \sin^2 \pi y)) \sin \pi y + 3 \sin \pi y - 4 \sin^3 \pi y$$

$$= 5 \sin \pi y + 2 \sin^3 \pi y = \sin \pi y (5 + 2 \sin^2 \pi y)$$

$$\text{Since } f(x) \geq 0 \Rightarrow \sin \pi y \geq 0$$

$$\Rightarrow \sin \pi\left(x - \frac{1}{4}\right) \geq 0 \Rightarrow \pi x - \frac{\pi}{4} \in [2n\pi, (2n+1)\pi]$$

$$x \in [0, 2] \Rightarrow \pi x - \frac{\pi}{4} \in \left[-\frac{\pi}{4}, \frac{7\pi}{4}\right]$$

From both of these,

$$\pi x - \frac{\pi}{4} \in [0, \pi]$$

$$\Rightarrow x \in \left[\frac{1}{4}, \frac{5}{4}\right]$$

$$\text{So, } \alpha = \frac{1}{4}, \beta = \frac{5}{4}$$

$$\beta - \alpha = \frac{5}{4} - \frac{1}{4} = 1$$

Q5. Let α and β be real numbers such that $-\frac{\pi}{4} < \beta < 0 < \alpha < \frac{\pi}{4}$. If $\sin(\alpha + \beta) = \frac{1}{3}$ and $\cos(\alpha - \beta) = \frac{2}{3}$, then the greatest integer of $\left(\frac{\sin \alpha}{\cos \beta} + \frac{\cos \beta}{\sin \alpha} + \frac{\cos \alpha}{\sin \beta} + \frac{\sin \beta}{\cos \alpha}\right)^2$ is _____.

Ans. 1

$$\begin{aligned} \text{Sol. } & \left(\frac{\sin \alpha}{\cos \beta} + \frac{\cos \alpha}{\sin \beta} + \frac{\cos \beta}{\sin \alpha} + \frac{\sin \beta}{\cos \alpha}\right)^2 \\ &= \left(\frac{\cos(\alpha - \beta)}{\sin \beta \cos \beta} + \frac{\cos(\alpha - \beta)}{\sin \alpha \cdot \cos \alpha}\right)^2 \\ &= \left(\frac{4}{3} \left\{ \frac{1}{\sin 2\beta} + \frac{1}{\sin 2\alpha} \right\}\right)^2 \\ &= \frac{16}{9} \left(\frac{2 \sin(\alpha + \beta) \cdot \cos(\alpha - \beta)}{\sin 2\alpha \cdot \sin 2\beta}\right)^2 \\ &= \frac{16}{9} \left(\frac{4 \cdot \frac{1}{3} \cdot \frac{2}{3}}{\cos(2\alpha - 2\beta) - \cos(2\alpha + 2\beta)}\right)^2 \\ &= \frac{16}{9} \left(\frac{\frac{8}{9}}{2\cos^2(\alpha - \beta) - 1 - 1 + 2\sin^2(\alpha + \beta)}\right)^2 \\ &= \frac{16}{9} \left(\frac{\frac{8}{9}}{\frac{8}{9} - 2 + \frac{2}{9}}\right)^2 \\ &= \frac{16}{9} \end{aligned}$$